

ROUTE SENSE

Know **where** rides are.
Know **when** they're available.



- SMARTER DECISIONS**
Find the best places and times to ride.
- REAL-TIME INSIGHTS**
Up-to-date community intelligence.
- CONNECTED JOURNEYS**
Know your ride. Know your signal.
- PRIVACY FIRST**
Your data stays protected.

1 THE PROBLEM

Knowing a transport route does not mean knowing whether a ride is actually available.

Informal public transportation can change depending on the time, location, demand, traffic, weather, and day of the week. Commuters may already know where vehicles usually pass, but they still face questions such as:

- “Can I actually get a ride here right now?”
- “Where is the better place to wait?”
- “How reliable is transportation at this hour?”
- “Will I still have mobile signal during the trip?”

Existing static route maps cannot fully answer these changing, real-world conditions.

2 BENEFICIARIES

PRIMARY USERS

- Students and daily commuters
- Workers and late-night commuters
- Delivery and gig-economy workers

SECONDARY BENEFICIARIES

- Schools and universities
- Transport cooperatives and associations
- Local government and transport planners
- Researchers
- Telecommunications and mobility planners

3 THE SOLUTION

RouteSense is an Android-based mobility intelligence platform that uses sensors already available inside passengers' smartphones to understand how informal transportation behaves over time.

- StopSense**
Learns patterns related to passenger movement, probable pickup areas, transport activity, and service availability.
- SignalTrail**
Measures cellular connectivity along frequently used transport corridors.

Instead of simply showing commuters where a route exists, RouteSense estimates:

- Where rides are likely available
- When they are available
- How reliable the estimate is
- How connected the journey may be

4 HOW IT WORKS

- Opt In**
The commuter voluntarily enables passive mobility sensing.
- Sense Movement**
The phone detects activity changes such as walking, waiting, and travelling in a vehicle.
- Collect Context**
GPS/GNSS, accelerometer, gyroscope, time, and cellular signal measurements provide mobility context.
- Analyze On-Device**
RouteSense estimates probable boarding areas, travel corridors, movement patterns, and connectivity conditions.
- Protect Privacy**
Detailed raw movement data is minimized and converted into anonymous or aggregated observations.
- Build Community Intelligence**
Observations from many trips reveal service patterns, pickup activity, reliability, and connectivity.
- Recommend**
RouteSense suggests transportation options based on current and historical conditions.

5 KEY FEATURES

- Service Probability**
Estimates how likely a commuter is to find transportation at a particular location and time.
- Smart Pickup Areas**
Identifies locations where successful transport activity frequently occurs.
- Opportunity Window**
Shows a realistic waiting range such as **Typical ride opportunity: 4-7 minutes** instead of pretending to know an exact vehicle arrival time.
- Time-Based Mobility**
Learns how transport availability changes between morning, afternoon, evening, weekdays, and weekends.
- Confidence & Freshness**
Every recommendation shows how much recent evidence supports it.
High Confidence • 31 observations at similar times • Last observation: 8 minutes ago
- Connectivity-Aware Routes**
Compares transportation choices based on both mobility and cellular connectivity.
- Route Pattern Discovery**
Detects recurring and emerging transport corridors from aggregated passenger movement.
- Offline Cached Insights**
Previously learned mobility information remains accessible during weak connectivity.
- No Driver App Required**
RouteSense learns from participating passenger smartphones rather than requiring every vehicle to join the platform.

6 NO DEDICATED GPS TRACKER OR DRIVER APPLICATION IS REQUIRED.

- Smartphone Sensor Fusion:** Combines multiple phone sensors to understand movement more accurately than GPS alone.
- Activity Recognition:** Identifies transitions such as: *Walking* → *Waiting* → *In Vehicle* → *Walking*.
- GNSS / Geospatial Analytics:** Analyzes movement trajectories and spatial patterns.
- Cellular Signal Intelligence:** Measures network conditions experienced during real passenger journeys.
- Context-Aware Computing:** Considers location, time, activity, and environmental context when interpreting mobility.
- On-Device Inference:** Processes sensitive movement information locally before sharing minimized results.
- Geospatial Clustering:** Finds recurring pickup areas, transport corridors, and mobility hotspots.
- Predictive Analytics:** Estimates service probability based on historical and recent observations.
- Privacy-Preserving Data Aggregation:** Builds community mobility intelligence without publishing individual travel histories.

PHONE HARDWARE USED

- GPS / GNSS
- Accelerometer
- Gyroscope
- Cellular Modem
- Compass / Magnetometer

8 EXAMPLE

Going to Visayan Village - 8:30 PM

- | OPTION A | Main Pickup Area | OPTION B | Alternate Pickup Area |
|-----------------------------------|---------------------------------------|-----------------------------------|--|
| Service Probability
88% | Typical Opportunity
3-6 min | Service Probability
74% | Typical Opportunity
6-10 min |
| ✓ Confidence: High | Connectivity: Moderate | ✓ Confidence: Moderate | Connectivity: Very Good |
| FASTEST CHOICE: OPTION A | | BEST CONNECTED: OPTION B | |

7 TECH STACK

- Mobile App
React Native + TypeScript
- Android Sensor Layer
Kotlin
- Local Storage
SQLite
- Maps
MapLibre + OSM
- Geospatial Backend
Supabase + PostgreSQL
- Geospatial Processing
PostGIS + H3
- Data Analytics
Python + pandas + scikit-learn
- Predictive Models
XGBoost / lightweight models

9 PROJECT TEAM

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CCE 106 — Application Development & Emerging Technologies | Student Researchers

10 REFERENCES & CITATIONS

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